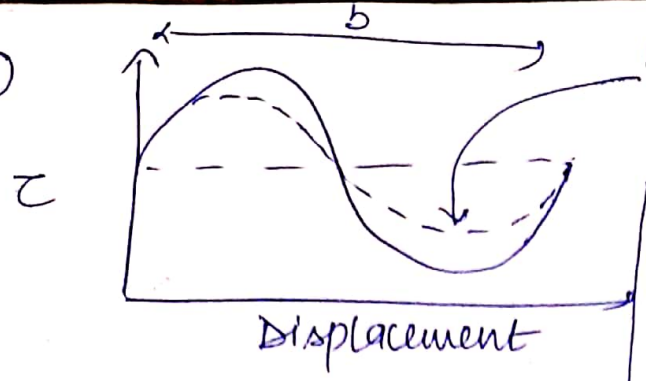
(1)



(2) $\tau = \tau_m \sin\left(\frac{2\pi x}{b}\right)$

$\tau = \tau_m\left(\frac{2\pi x}{b}\right)$

also $\tau = G\gamma = G\frac{x}{a}$

$G\frac{x}{a} = \tau_m \frac{2\pi x}{b}$

if b n a

$$\boxed{\tau_m = \frac{G}{2\pi}}$$

(3) G_{metal} - 20-150 GPa

τ_{pr} = 30-30 GPa

τ_{th} = 0.5-10 MPa

(4) Slip plane - higher planar δ
Slip direction - higher linear δ

(5) Preferred crystallographic direction - slip direction

preferred dislocation movement - slip plane

(6) BCC & FCC > HCP
(ductile & slip systems)

(7) - FCC Cu, Al, Ni

	S.P	S.D	Total
	{111}	⟨$\bar{1}$10⟩	12
- BCC			
α-Fe	{110}	⟨$\bar{1}$11⟩	12
α-Fe	{211}	⟨$\bar{1}$11⟩	12
α-Fe	{321}	⟨$\bar{1}$11⟩	24
- HCP - Ti, Mg, Zn			
	{0001}	⟨11$\bar{2}$0⟩	3
	{10$\bar{1}$0}	⟨1$\bar{1}$20⟩	3
	{10$\bar{1}$1}	⟨11$\bar{2}$0⟩	6

— = predominant density

(8) Since not so closed structure, high shearing stress required for slip, slip lines are mostly wavy only because it happens in one direction only.

(9) Flow stresses - anisotropic
Dislocations only glide when shear stresses there.

(10) In uniaxial stress, shear comp. of applied test determined on slip plane.

(11) Resolved shear stress - comp of applied stress & mag. depend on orientation of slip plane & direction within that plane.

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(12) $A = A_s \cos\phi$

$$\tau_R = \frac{P\cos\lambda}{A_s} = \frac{P\cos\lambda\cos\phi}{A}$$

along slip plane)

$$= \sigma \underbrace{\cos\lambda\cos\phi}_{\text{Schmid factor}}$$

3) Diff. orientation with load axis. one slip favorable. RSS.

4) slip on favorable plane occurs when RSS reaches critical value τ_{CRSS}.

↓

min shear stress to initiate slip & when yielding starts.

15) $\tau_{R\,max} = \tau_{CRSS}$

Y. stress ↓ when $\phi = \lambda = 45°$

so $2\tau_{CRSS} = \sigma_y$

6) $\lambda = 90° / \phi = 90° \rightarrow RSS = 0$

7) $[\,] \perp (\,)$ $u = h, v = k, w = l$

$[\,] \parallel (\,)$ $hu + vl + wk = 0$

normal $h_1h_2 + k_1k_2 + l_1l_2 = 0$

direction normal $u_1u_2 + v_1v_2 + w_1w_2 = 0$

example → $\dfrac{h_1h_2 + k_1k_2 + l_1l_2}{\sqrt{h_1^2+k_1^2+l_1^2}\sqrt{h_2^2+k_2^2+l_2^2}}$

(18) Polycrystalline metals are stronger than their single crystal equivalents.

(19) Mechanical twinning

↓

BCC & HCP, at low temp and high loading rate, & slip process is restricted.

(20) Shear deformation in twinning homogenous.

(21) Bulk plastic deformation from twinning is normally small relative to slip.

(22) During twinning crystallographic reorientation may result in new slip system

(23) Min. τ to move a dislocation - Peierls stress.

↓

function of structure of core

(a) (width) bonding structure (b) of crystal

(24) width of dislocation basis for ease of motion of dislocation.

(25) Larger width → small atomic adjustments (relaxed)

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(26) $\tau_{PN} = Ge^{-\left(\frac{2\pi w}{b}\right)}$

(27) Covalent → small w
so high τ_{PN}.
fail by brittle.

Metallic → low τ_{PN}
Ionic - high b.

(28) $\frac{d\epsilon}{dt} \propto e^{-\left(\frac{Q}{KT}\right)}$

↳ random & no plastic deformation.

(29) $\frac{d\epsilon}{dt} \propto e^{-\left(\frac{(\tau_{PN}-\tau_{app})v}{KT}\right)}$

activation volume.

(30) Dislocation density
Annealed - 10^4-10^6 mm^{-2}
Heavily deformed - 10^8-10^9 mm^{-2}

(31) dislocations multiply as large dislocation density in cold worked metals.

→ frank-read source.

(32) Chemical attack due to strain field around site.
Etchant forms pit.
↳ 500Å apart with DD 10^8 mm^{-2}

(33) photolytic silver → Hedge & Mitchell for decoration of dislocation.

(34) TEM
100 nm from 1mm using electro-polishing.

(35) Stiffness, strength, hardness, ductility & toughness

SSHDT

(36) Compressive when material behavior under large plastic strain is desired.

(37) Tangent modulus → at certain stress

Secant modulus → slope of secant from origin to point.

(38) $\nu = 0.25$-0.35
$E = 2G(1+\nu)$
$E = 3K(1-2\nu)$

(39) Strain about 0.005 for elastic deformation.

(40) Crystalline → slip
Amorphous → viscous flow mechanism.

(41) Offset → 0.002

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(12) $\varepsilon_n = \frac{(L-L_0)}{L_0}$

$\sigma_n = P/A_0$

$$\boxed{\varepsilon_t = \ln(1+\varepsilon_n)}$$

$$\boxed{\sigma_t = \sigma_n(1+\varepsilon_n)} \quad \left[A_i = \frac{L_0 A_0}{L_i}\right]$$

(13) $\sigma_T = K\varepsilon_t^n$ → strain hardening exponent.

(14) Toughness → energy to break unit volume of material.
Area under σ-ε curve.

(15) Resilience $U_r = \int_0^{\varepsilon_y} \sigma \, d\varepsilon$

$U_r \approx \frac{1}{2}\sigma_y \varepsilon_y$

(16) Vicker → 136° (diamond pyramid)
Knoop →

(17) Jog → out of slip plane sharp break.
Kink → in plane.

(18) Two ⊥ edge dislocation & two ∥ b result in two jogs.

(19) Screw & edge
↓ ↓
Kink Jog.

screw & screw
↓
jog.

(20) pure edge → no movement of jog.

in screw
↳ jog can move is by slip along axis of screw.
↳ climb.
↳ thermally activated
→ so temp. dependent jogged screw dislocation.

Stiffness - material's resistance to elastic deformation

Tension test
Load - stroke
↓
F